5 Related Literature Review

Studies of pumped-storage hydropower differ in what they value, which uncertainties they represent, which markets they include and how they solve the resulting planning problem. This chapter reviews these four themes in turn. Section 5.1 describes how the review is limited and how the literature was found. Section 5.2 reviews how the value of pumped storage and of its capacities has been assessed, and Section 5.3 how uncertainty and price dynamics have been represented. Section 5.4 reviews participation in energy and reserve markets, and Section 5.5 the solution methods used and the physical detail they allow. Finally, Section 5.6 identifies the research gap and positions this study. Table 3 compares the reviewed studies.

5.1 Scope and Search

The review covers the operational scheduling and economic valuation of pumped-storage hydropower, with emphasis on studies that link short-term operation to the long-term management of stored water. Studies of conventional reservoir hydropower are included where they show how uncertainty, reserve markets or plant detail affect water values, since a reservoir plant faces the same storage decision without the option to pump. System studies and studies of hybrid island systems are included where they assess pumped-storage capacity, because they show how the value of a capacity depends on the objective of the study. Studies of price dynamics and storage value are included because the research question concerns price volatility and persistence.

TODO (authors): describe the search you actually carried out, in three to five sentences: databases (e.g. Google Scholar, Scopus, Oria), search strings, inclusion criteria (peer-reviewed, title and abstract within scope) and whether further studies were found by following reference lists (backward snowballing, as Bore and Brensholm describe in their Section 4.1). The search log (`literature/search-log.md`) is still empty, so nothing may be stated here before it has been logged.

5.2 Valuation of Pumped Storage and Its Capacities

The value of pumped storage depends on whose value is measured. System studies measure the change in total surplus, whereas owner studies measure the operating profit of one plant. The two can point in opposite directions. Henden et al. (2016) add between 0 and 14 250 MW of new pumped storage in southern Norway to a model of the Northern European power system in 2030. Only about 950 MW maximises socio-economic surplus, and owners need total exchange capacity of more than about 11 000 MW before the investment pays off. With high transmission capacity, a large pumped-storage fleet runs often but earns almost nothing, because its own operation removes the price differences it trades on. Dallinger et al. (2019)

find the same effect in Central Europe for 2030: the average price spread captured by pumped storage falls as Austrian storage capacity grows. These studies show that the value of storage is not a fixed property of the plant but falls as more storage is built. They value energy trading only, which the authors of the first study expect to understate the value of pumped storage. For a single owner, the result means that the price-taker assumption holds only if the plant is small relative to the market it trades in.

Helseth et al. (2018) assess an expansion of the 200 MW Duge pumped-storage plant in southern Norway by up to 2000 MW in a detailed model of the Nordic system. Their detailed model gives a smoother, but lower, marginal value of the additional capacity than an aggregated model, because it represents the constraints of the surrounding watercourse. The paper reports the result as a change in socio-economic surplus and does not state whether pump and turbine capacity are expanded separately. It therefore shows that the representation of the plant changes the value of capacity, but not which capacity is worth expanding.

Owner studies that add a pump to an existing cascade report small gains. Ak et al. (2019) convert a three-reservoir Turkish cascade into a pumped-storage system and find that net revenue rises by 2.9 to 10.4%, 6.0% on average, across five historical price years. Turbine revenue rises by 23 to 32%, but most of this gain is offset by the cost of pumping. Alic et al. (2023) retrofit a 38 MW pump to the upper plant of a Norwegian cascade and find a median revenue gain of only 2.35%. Schüller et al. (2024) value a new 90 MW plant with pumping in the Sira-Kvina system and find that net income rises by 2.9% in their medium-term model but by 1.9% when the operation is simulated with hourly detail, because short-term restrictions, such as a minimum head for pumping, reduce the use of the new plant. All three studies use the energy market only, and each evaluates a single pump size. In Alic et al. (2023), the pump also has the same power as the turbine, so the effect of pump size cannot be separated from that of adding a pump at all.

Studies that include reserve markets find larger and more varied gains from a change in configuration. Helseth et al. (2017) compare two reinvestment alternatives of 24.8 MW for a Norwegian reservoir plant. Selling primary reserves raises expected operating profit by 7.3% for a configuration with a large Francis and a small Pelton unit and by 10.8% for one with two Pelton units, since Pelton units can run at low output and leave more capacity for reserves. Chazarra et al. (2017) compare a conventional closed-loop plant with two plants that can pump and generate at the same time. The configuration with ternary units earns 141 to 166% more than the conventional plant in two representative weeks, whereas the configuration with reversible units earns only 2.8 to 3.5% more. In both studies, the configuration that gives the most flexible operating range wins because reserve capacity is paid. Neither varies the size of the reservoir. Chazarra et al. (2017) note that the flexible plant uses only a small part of its storage and suggest, without testing it, that this could affect reservoir sizing.

Bore and Brensholm (2026) come closest to valuing capacity for an owner under uncertainty. They add 30, 60 and 90 MW of generating capacity to a conventional plant in southern Norway. Selling reserve capacity raises revenue by 25.8% compared with energy trading alone

for the plant without an upgrade, and by 57.5% with the 90 MW upgrade (Bore et al., 2026). Energy revenue hardly changes with the upgrade, because the plant has no more water to sell. The extra capacity is therefore paid mainly through reserve markets, and the marginal benefit of each step falls (Bore and Brensholm, 2026). Their upgrades change only generating power, and the plant has no pump.

The value of reservoir capacity has been examined mainly outside electricity markets. Zisos et al. (2023) size the upper reservoir of a seawater pumped-storage plant on a Greek island, with pump and turbine both fixed at 6 MW. The optimal storage is 315 195 m^3 for the deterministic case and about 330 000 m^3 on average over 100 weather and demand scenarios, less than a third of the 1.1 Mm^3 of an earlier design. The plant follows a fixed rule that pumps all surplus and covers all deficits, and its value comes from reduced unmet demand rather than from prices. The study shows that the reservoir size matters, but with fixed pump and turbine power and without prices it cannot show how the three capacities interact.

The reviewed studies thus vary one capacity at a time: a pump is added, a generator is upgraded or replaced, or a reservoir is sized. None varies pumping power, generating power and reservoir capacity jointly for a plant that trades in markets. This matters because one capacity may limit another. The finding of Bore and Brensholm (2026) that extra generating power earns little energy revenue when water is scarce is one example; a larger pump adds little when the reservoir is already full.

5.3 Uncertainty and Price Dynamics

Prices and inflow are unknown when the owner decides whether to store water, and the value of storage depends on how they develop. The reviewed studies differ in which inputs they treat as uncertain, how they describe them and how much the decision maker is assumed to know.

Several studies describe the future through historical or simulated years and optimise each year with full knowledge of it. Ak et al. (2019) solve the cascade problem for each of five historical price years, so each result assumes that the whole year is known in advance. Chazarra et al. (2017) schedule each day with known prices and reserve activation, and Naval and Yusta (2025) treat the 2023 Spanish day-ahead prices as known for a whole year while treating only wind as uncertain. The reported revenues are therefore upper bounds on what an operator can earn, and the authors of the second study state this themselves. Helseth et al. (2018) let each week's decision be informed by 19 weather and price scenarios drawn from historical years. All of these scenarios are revealed at once after the first week, so the plan for later weeks is made with knowledge of each scenario.

Other studies let information arrive week by week. Alic et al. (2023) cluster weekly price and inflow observations into five joint states per week and estimate the probabilities of moving between them. Helseth et al. (2017) and Helseth et al. (2016) build weekly states for energy and reserve prices, and Helseth et al. (2016) model inflow separately. Bore and Brensholm (2026) use three joint price and inflow states per week. Such a description keeps the dependence between consecutive weeks, but each week's outcome is represented by a handful

of typical weeks. In Alic et al. (2023), weekly average prices are spread over the week with a fixed profile, so the variation of prices within the week, which a pump exploits, is the same in every state. This may explain part of their small gain from pumping, although the authors do not say so.

How the dependence between prices and inflow is described appears to matter less for revenue than for operation. Kleiven et al. (2023) find that ignoring the negative correlation between Norwegian prices and inflow lowers water values and leads to a higher reservoir in autumn, but costs only 0.17 to 0.30% of expected revenue. Pérez-Díaz et al. (2020) likewise find that the correlation between energy and reserve prices has little effect on water values for a Spanish reservoir plant. These results concern conventional plants and say little about whether the same holds when prices also drive pumping.

The information available within the week is a separate question. Abgottspon and Andersson (2014) compare four ways of describing the week for a Swiss pumped-storage plant. Assuming that the whole week is known overvalues water. Representing the week as uncertain raises expected profit and its conditional value at risk by about 10% when weekly secondary reserves are included, but gives no advantage without them. Alic et al. (2023) show a related effect of time resolution: when operation is simulated day by day instead of week by week, revenue falls by 24% in the first week in which an environmental restriction on the reservoir applies. Both results suggest that a known week is a reasonable simplification for energy trading, but less so when reserves or state-dependent restrictions are included.

Few of the reviewed studies examine how the behaviour of prices over time affects the value of storage, although this is what a pump and a reservoir earn from. Electricity prices in the Nordic market are both volatile and persistent. Haldrup and Nielsen (2006) find that hourly Nord Pool prices in 2000–2003 revert to their mean only slowly, with the strongest persistence in the Norwegian price areas, which they link to the long memory of reservoirs and inflow. Regimes with and without transmission congestion last on average from about 5 to 77 hours. Mo et al. (2001) note that a description in which next week's price depends only on this week's state forgets the past quickly, so price uncertainty about prices far into the future needs an additional model. Section ?? defines volatility and persistence as used in this study.

The link between price variation and storage value has mainly been examined with known prices. Ak et al. (2019) find the largest gain from pumping in the price year with the largest variation within the day, but the relation is not one-to-one, because changes from month to month and head also matter. Sioshansi et al. (2009) find that, with known prices in the PJM market, the first four hours of storage earn more than half of the arbitrage value, eight hours about 85% and twenty hours about 95%. Bradbury et al. (2014) show, for seven US markets, that greater price volatility raises the return on storage, but that the most profitable ratio of storage to power is set mainly by efficiency and by how the cost is split between power and storage, not by volatility. Löhndorf and Wozabal (2023) report that coordinating day-ahead and intraday trading is worth 10.0 to 29.1% for a pumped-storage plant but only 1.0 to 1.6% for a reservoir plant without a pump, which suggests that short-term price variation matters more for a plant that can pump. None of these studies varies volatility and persistence

separately, and none relates them to the value of pump, turbine and reservoir capacity under uncertainty. Since a pump and a turbine earn from short price differences, whereas a reservoir earns from differences that persist over weeks or seasons, the two properties may favour different capacities.

5.4 Market Participation

The markets included decide which revenues a study can report. Several of the reviewed owner studies include only day-ahead energy (Ak et al., 2019; Alic et al., 2023; Schüller et al., 2024), as do the system studies of Henden et al. (2016) and Helseth et al. (2018). Abgottspon and Andersson (2013) add forward contracts for a Swiss plant with 240 MW of generating power, 46 MW of pumping power and about 200 GWh of storage, which they model as a producer that affects prices. Forward trading raises simulated annual profit from about 20 to 24 million euro. Forward contracts spread the sales of the same energy over time, however, and do not pay for flexibility as reserves do.

For conventional reservoir plants, reserve sales consistently raise profit and change how the reservoir is used. Helseth et al. (2016) sell symmetric reserve capacity, traded the week before energy, from a Norwegian watercourse of seven plants. A large part of the reserve profit comes without changing the generation schedule, because the plants are most efficient below full output and therefore have spare capacity both up and down. With reserves, the large reservoirs are kept higher until the spring flood and then used more to keep generators running. Helseth et al. (2017) find the same pattern for a single reservoir, together with water values that fall faster as the reservoir fills, because capacity held for reserves limits how fast water can be discharged. Pérez-Díaz et al. (2020) find that including the secondary reserve market when computing water values raises the profit of a Spanish reservoir plant by about 4.5%. Bore and Brensholm (2026) add intraday energy and four Nordic reserve products and find that reserve revenues provide a hedge, since they make up a larger share of revenue in periods with high inflow and low energy prices.

A pump extends what a plant can offer in reserve markets. Dallinger et al. (2019) show that adding pump-turbines to an Austrian reservoir plant extends its operating range from minimum load up to full output to between about −70% and 100% of turbine capacity. The plant can then offer downward reserve without generating, and its ability to deliver reserves no longer depends on natural inflow. Chazarra et al. (2017) find that a closed-loop plant earns most of its income from secondary reserve capacity, and that the day-ahead market becomes a means of enabling reserve sales. A conventional plant cannot offer reserves while pumping, which is why the configuration that pumps and generates at the same time earns much more. Abgottspon and Andersson (2014) include weekly secondary reserves for a Swiss pumped-storage plant and find that they raise expected profit for all four models compared.

The treatment of reserve activation differs widely. Chazarra et al. (2017) include the energy delivered when reserves are activated, based on historical activation, and show that storage limits restrict reserve offers: the conventional plant does not offer reserves in an hour when the expected activation of about 85% could empty the upper reservoir. Dallinger et al. (2019)

simulate activations every quarter hour, but offer capacity at a price of zero, so their reserve revenue comes from activated energy only. The other reviewed studies pay for capacity and leave activation out (Helseth et al., 2016; Helseth et al., 2017; Abgottspon and Andersson, 2014; Bore and Brensholm, 2026). Leaving activation out omits its energy settlement and its effect on stored water, which matters most for a plant with a small reservoir. Most evidence on reserve markets and water values concerns conventional plants. For pumped storage, only Abgottspon and Andersson (2014) combine reserves with water values over a year, for a single plant whose capacities are not reported.

5.5 Solution Methods

The studies above combine a long horizon, uncertain inputs, several markets and different levels of plant detail. No method handles all of these equally well, and the physical detail a study can keep depends on the method it uses. This section reviews the methods applied, whose principles are described in Sections ??, ?? and ??.

Deterministic optimisation and simulation

Deterministic models solve one problem with given inputs and can therefore include nonlinear and discrete plant features. Chazarra et al. (2017) use a mixed-integer linear model to represent unit modes and simultaneous pumping and generation, solved one day at a time with the same reservoir level at the start and end of each day. Ak et al. (2019) use a nonlinear model with head-dependent generation and monthly decisions, in which hourly prices enter only through representative price curves. Naval and Yusta (2025) solve a mixed-integer model for a hybrid wind, solar and pumped-storage system over a year, with chance constraints on how much of the demand renewables cover, and Zisos et al. (2023) combine simulation with a heuristic search over reservoir sizes. These models give results with full knowledge of the inputs, so they do not provide a value of stored water that an operator could use without knowing the future. Where the horizon is longer than a day, the use of the reservoir is fixed by end conditions or monthly rules rather than by a water value.

Two-stage stochastic programming in a rolling horizon

Helseth et al. (2018) solve, every week, a two-stage stochastic linear programme in which the first stage is the current week and the second stage contains one scenario for each of 19 historical years, and then move one week forward. This keeps all 1265 hydropower modules of the Nordic system without aggregation and uses historical weather directly. The price is a long run time, 50 hours for a transmission case and 156 hours for the pumped-storage case, and the plan for later weeks is made with knowledge of each scenario. The value of water at the end of the scenarios is taken from another model.

Stochastic dynamic programming

Stochastic dynamic programming computes water values on a grid of reservoir levels for each stage and state, and places no convexity requirement on the stage problems. Alic et al. (2023) choose it because their environmental restrictions depend on the reservoir level and are not convex, even though they note that stochastic dual dynamic programming is more efficient for

medium- and long-term scheduling. Helseth et al. (2017) solve weekly mixed-integer problems with head-dependent generation and unit commitment. A linear version of their model overestimates profit with reserve sales by 2.3 to 5.2 percentage points, mainly because it lets the plant sell reserves below its minimum output. Water values from the linear model, used in a simulation with the mixed-integer model, lose 0.6 to 1.0 percentage points. The cost of this detail is the grid: Alic et al. (2023) need hours to solve two reservoirs with 25 grid points each, and the reviewed applications concern one plant or a small cascade.

Stochastic dual dynamic programming

Stochastic dual dynamic programming (Pereira and Pinto, 1991) avoids the grid by approximating the expected future profit with linear cuts that are refined iteratively, and has become the standard method for medium- and long-term hydropower scheduling (Füllner and Rebennack, 2025). Its convergence results require convex, typically linear, stage problems (Shapiro, 2011). Helseth et al. (2013) apply it to the Icelandic hydrothermal system with a 100 MW pumped-storage plant, and describe the exogenous price by weekly states between which the cuts are computed separately, which keeps the problem convex. Their linear approximation of pump start-up costs reduces switching from generation to pumping from 6.54 to 0.12% of the time periods, but the pump still runs below its minimum output, so the approximation is imperfect. Helseth et al. (2016) use the same combination of price states and cuts to include uncertain reserve prices, and note that the linear plant description may cause significant errors when reserves are sold.

Several studies relax the linearity. Hjelmeland et al. (2019) include head-dependent generation and use strengthened cuts. These cuts give a better operating policy than ordinary cuts at about five times the run time, but with uncertain prices the gap between the bounds is still 36%. Helseth (2024) represent the detailed Norwegian hydropower system in an aggregated model through feasibility cuts computed from the detailed system, building on Helseth and Mo (2023). On 30 historical years this lowers total cost by 7.2% on average compared with plain aggregation, at 2.7 times the run time. These extensions bring the plant description closer to that of stochastic dynamic programming, but at a computational cost.

The method has also been used for single pumped-storage plants. Abgottspon and Andersson (2016) schedule a Swiss system of ten reservoirs, with 1125 MW of turbines and 424 MW of pumps, by using only the four large reservoirs as states between days and keeping the six small reservoirs inside hourly problems within each day. Their model reaches a gap of 0.5%, but the reported case is deterministic. Schüller et al. (2024) use water values from a medium-term model of this type to value a pumping investment. Bore and Brensholm (2026) use the method with three price and inflow states per week for a conventional plant. In their model, each iteration samples a single path through the weeks, the algorithm stops after a fixed 50 iterations, and no bounds are reported. They also find that the water value does not fall as the reservoir approaches full, which they attribute to the few reservoir levels the algorithm visits.

The methods thus trade plant detail against the size of the problem and the description of uncertainty. Deterministic models keep the most detail but give no water values under uncertainty, and two-stage models add scenarios but reveal them all at once. Stochastic

dynamic programming keeps detail and sequential decisions for small systems, whereas stochastic dual dynamic programming handles more states and finer resolution if the stage problems are linear. A study that compares many configurations under several price cases needs a method whose run time allows many solutions, and the results of Helseth et al. (2017) indicate what a linear plant description costs.

5.6 Research Gap and Positioning of This Study

Table 3 summarises the reviewed studies. The literature shows that reserve markets change both the profit and the water values of a hydropower plant, that pumping extends what a plant can offer in these markets, and that the value of storage falls as more storage is built. It also shows that the gains from adding a pump to an energy-only plant are small, from about 2 to 10%, while flexible configurations in reserve markets can earn much more.

The closest studies each cover part of the question of which capacities create value. Bore and Brensholm (2026) value generating-capacity upgrades under uncertainty in several markets, but for a plant without a pump. Alic et al. (2023) and Schüller et al. (2024) value a single pump in the energy market, Helseth et al. (2017) and Chazarra et al. (2017) compare unit configurations of given size, and Zisos et al. (2023) size a reservoir with fixed pump and turbine and no market. Helseth et al. (2018) expand a pumped-storage plant from a system perspective without separating its capacities. No reviewed study varies pumping power, generating power and reservoir capacity jointly for a price-taking owner in energy and reserve markets under uncertainty. In addition, the effect of price behaviour on storage value has been examined only with known prices, and never with volatility and persistence varied separately.

This study therefore asks how pumping power, generating power and reservoir capacity, individually and in combination, affect the long-term operating value of a price-taking owner that trades in day-ahead and reserve-capacity markets, and how the ranking of these capacities changes with price volatility and persistence. Following the reviewed applications to reservoir hydropower, the problem is solved with stochastic dual dynamic programming, with weekly stages and uncertain inflow and day-ahead prices. The last row of Table 3 places the study among the reviewed literature. The contribution is a comparison of capacity combinations under uncertainty, not a claim that stochastic scheduling, reserve participation or storage sizing is new on its own.

Table 3: Reviewed studies compared by perspective, uncertainty, markets, capacity question and solution method, and the position of this study.

Study	Perspective and asset	Uncertainty and information	Markets	Capacity question	Solution method
Helseth et al. (2013)	System; Icelandic hydrothermal system with a 100 MW PHS	Inflow, wind and exogenous price; weekly stages, week known	No market products; exogenous price	Operation; effect of grid and pump start-up costs	SDDP with price states
Abgottspon and Andersson (2013)	Price-making owner; Swiss PHS, 240/46 MW, 200 GWh	Inflow; one year	DA and forward contracts	Operation of the given plant	SDP
Abgottspon and Andersson (2014)	Owner; Swiss PHS, capacities not reported	Inflow and prices; four descriptions of the week compared	DA and weekly secondary reserve capacity	Operation of the given plant	SDP with hourly within-week problems
Helseth et al. (2016)	Price-taking owner; Norwegian watercourse, seven plants	Inflow, energy and reserve prices; weekly stages	DA and reserve capacity traded the week before	Operation; water values with reserves	SDDP with price states
Helseth et al. (2017)	Price-taking owner; Norwegian reservoir plant, no pump	Inflow, energy and reserve prices; weekly states	DA and primary reserve capacity	Two reinvestment alternatives of 24.8 MW	SDP with weekly MILP
Chazarra et al. (2017)	Price-taking owner; closed-loop PHS	None; known prices and activation, daily	DA, secondary reserve capacity and activation energy	Three unit configurations, simultaneous pumping and generation	MILP
Helseth et al. (2018)	System; Nordic power system	Weather and exogenous prices; all scenarios revealed after the first week	No market products	Duge PHS expanded by up to 2000 MW	Two-stage stochastic LP in a rolling horizon
Ak et al. (2019)	Price-taking owner; Turkish cascade	None; each historical price year known	DA energy	Existing cascade versus pumped-storage operation	NLP with monthly decisions
Dallinger et al. (2019)	System and owner; Central Europe and an Austrian plant	None; one historical year known	DA and aFRR (activation energy only)	Adding pump-turbines; lower minimum load	Dispatch model and MILP
Alic et al. (2023)	Price-taking owner; Norwegian cascade	Inflow and price; weekly states	Energy	Adding a 38 MW pump	SDP
Zisos et al. (2023)	Island system; seawater PHS	Wind, demand and power curve; 100 scenarios	None	Reservoir and solar sizing; pump and turbine fixed at 6 MW	Simulation with heuristic search
Schüller et al. (2024)	Price-taking owner; Sira-Kvina system	Inflow and price	Energy	A new 90 MW plant with pumping	SDDP-based water values, hourly simulation
Naval and Yusta (2025)	Owner of a hybrid system; closed-loop PHS	Wind; prices known for the year	DA energy	Operation of the given system	MILP with chance constraints
Bore and Brensholm (2026)	Price-taking owner; Norwegian reservoir plant, no pump	Inflow, DA price; capacity prices at three levels; week known	DA, ID, FCR-N, FCR-D and aFRR capacity; activation excluded	Generating-power upgrades of 30, 60 and 90 MW	SDDP with price and inflow states
This study	**Price-taking owner; pumped storage with natural inflow**	**Inflow and DA price; weekly stages, week known; varied price volatility and persistence**	**DA and reserve capacity; activation excluded**	**Combinations of pumping power, generating power and reservoir capacity**	**SDDP with price and inflow states**

SDP: stochastic dynamic programming; SDDP: stochastic dual dynamic programming; LP: linear programming; MILP: mixed-integer linear programming; NLP: nonlinear programming; PHS: pumped-storage hydropower; DA: day-ahead; ID: intraday; FCR-N/FCR-D: frequency containment reserve for normal operation/disturbances; aFRR: automatic frequency restoration reserve. Studies cited only for price dynamics or as general methodological references are not included.

References

Abgottspon, Hubert and Andersson, Göran (2013). Stochastic Scheduling for a Price-Maker Hydro Producer Considering Forward Trading. *2013 IEEE Grenoble Conference*, 1–6. doi: 10.1109/PTC.2013.6652304

Abgottspon, Hubert and Andersson, Göran (2014). Medium-Term Optimization of Pumped Hydro Storage with Stochastic Intrastage Subproblems. *Proceedings of the 18th Power Systems Computation Conference*. doi: 10.1109/PSCC.2014.7038352

Abgottspon, Hubert and Andersson, Göran (2016). Multi-Horizon Modeling in Hydro Power Planning. *Energy Procedia* 87, 2–10. doi: 10.1016/j.egypro.2015.12.351

Ak, Mümtaz, Kentel, Elcin and Savasaneril, Secil (2019). Quantifying the Revenue Gain of Operating a Cascade Hydropower Plant System as a Pumped-Storage Hydropower System. *Renewable Energy* 139, 739–752. doi: 10.1016/j.renene.2019.02.118

Alic, Asja, Schäffer, Linn Emelie, Toffolon, Marco and Trovato, Vincenzo (2023). Optimal Price-Based Scheduling of a Pumped-Storage Hydropower Plant Considering Environmental Constraints. *Energy Systems*. doi: 10.1007/s12667-023-00614-y

Bore, Martin and Brensholm, Sigurd Skogan (2026). Assessing the Value of Hydropower Capacity Upgrades: Using a Multi-Market Stochastic Optimization Framework. *Norwegian University of Science and Technology*. Master's thesis.

Bore, Martin and others (2026). Economic Assessment of Hydropower Capacity Expansions and Multi-Market Participation Using a Multi-Stage Stochastic Framework. *2026 22nd International Conference on the European Energy Market (EEM)*. doi: 10.1109/EEM68581.2026.11589591

Bradbury, Kyle, Pratson, Lincoln and Patiño-Echeverri, Dalia (2014). Economic Viability of Energy Storage Systems Based on Price Arbitrage Potential in Real-Time U.S. Electricity Markets. *Applied Energy* 114, 512–519. doi: 10.1016/j.apenergy.2013.10.010

Chazarra, Manuel, Pérez-Díaz, Juan I. and García-González, Javier (2017). Optimal Energy and Reserve Scheduling of Pumped-Storage Power Plants Considering Hydraulic Short-Circuit Operation. *IEEE Transactions on Power Systems* 32(1), 344–353. doi: 10.1109/TPWRS.2016.2545740

Dallinger, Bettina, Schwabeneder, Daniel, Lettner, Georg and Auer, Hans (2019). Socio-Economic Benefit and Profitability Analyses of Austrian Hydro Storage Power Plants Supporting Increasing Renewable Electricity Generation in Central Europe. *Renewable and Sustainable Energy Reviews* 107, 482–496. doi: 10.1016/j.rser.2019.03.027

Füllner, Christian and Rebennack, Steffen (2025). Stochastic Dual Dynamic Programming and Its Variants: A Review. *SIAM Review* 67(3), 415–539. doi: 10.1137/23M1575093

Haldrup, Niels and Nielsen, Morten Ørregaard (2006). A Regime Switching Long Memory Model for Electricity Prices. *Journal of Econometrics* 135(1--2), 349–376. doi: 10.1016/j.jeconom.2005.07.021

Helseth, Arild (2024). Approximating Hydropower Systems by Feasibility Spaces in Stochastic Dual Dynamic Programming. *Electric Power Systems Research* 234, 110615. doi: 10.1016/j.epsr.2024.110615

Helseth, Arild and Mo, Birger (2023). Hydropower Aggregation by Spatial Decomposition—An SDDP Approach. *IEEE Transactions on Sustainable Energy* 14(1), 381–392. doi: 10.1109/tste.2022.3214497

Helseth, Arild, Gjelsvik, Anders, Mo, Birger and Linnet, Úlfar (2013). A Model for Optimal Scheduling of Hydro Thermal Systems Including Pumped-Storage and Wind Power. *IET Generation, Transmission and Distribution* 7(12), 1426–1434. doi: 10.1049/iet-gtd.2012.0639

Helseth, Arild, Fodstad, Marte and Mo, Birger (2016). Optimal Medium-Term Hydropower Scheduling Considering Energy and Reserve Capacity Markets. *IEEE Transactions on Sustainable Energy* 7(3), 934–942. doi: 10.1109/TSTE.2015.2509447

Helseth, Arild, Fodstad, Marte, Askeland, Magnus, Mo, Birger, Nilsen, Odd Bjarte, Pérez-Díaz, Juan I., Chazarra, Manuel and Guisández, Ignacio (2017). Assessing Hydropower Operational Profitability Considering Energy and Reserve Markets. *IET Renewable Power Generation* 11(13), 1640–1647. doi: 10.1049/iet-rpg.2017.0407

Helseth, Arild, Mo, Birger, Henden, Arild Lote and Warland, Geir (2018). Detailed Long-Term Hydro-Thermal Scheduling for Expansion Planning in the Nordic Power System. *IET Generation, Transmission and Distribution* 12(2), 441–447. doi: 10.1049/iet-gtd.2017.0903

Henden, Arild Lote, Doorman, Gerard and Helseth, Arild (2016). Economic Analysis of Large-Scale Pumped Storage Plants in Norway. *Energy Procedia* 87, 116–123. doi: 10.1016/j.egypro.2015.12.340

Hjelmeland, Martin N., Helseth, Arild and Korpås, Magnus (2019). Medium-Term Hydropower Scheduling with Variable Head under Inflow, Energy and Reserve Capacity Price Uncertainty. *Energies* 12(1), 189. doi: 10.3390/en12010189

Kleiven, Andreas, Risanger, Simon and Fleten, Stein-Erik (2023). Co-Movements between Forward Prices and Resource Availability in Hydro-Dominated Electricity Markets. *Energy Systems*. doi: 10.1007/s12667-023-00605-z

Löhndorf, Nils and Wozabal, David (2023). The Value of Coordination in Multimarket Bidding of Grid Energy Storage. *Operations Research* 71(1), 1–22. doi: 10.1287/opre.2021.2247

Mo, B., Gjelsvik, A., Grundt, A. and Kåresen, K. (2001). Optimisation of Hydropower Operation in a Liberalised Market with Focus on Price Modelling. *Proceedings of the IEEE Porto Power Tech*.

Naval, Natalia and Yusta, Jose M. (2025). Optimization of Pumped Hydro Storage Management with Renewable Generation under Wind Uncertainty Using Chance-Constrained Programming. *Sustainable Energy, Grids and Networks* 43, 101889. doi: 10.1016/j.segan.2025.101889

Pereira, M. V. F. and Pinto, L. M. V. G. (1991). Multi-Stage Stochastic Optimization Applied to Energy Planning. *Mathematical Programming* 52, 359–375. doi: 10.1007/BF01582895

Pérez-Díaz, Juan I., Guisández, Ignacio, Chazarra, Manuel and Helseth, Arild (2020). Medium-Term Scheduling of a Hydropower Plant Participating as a Price-Maker in the Automatic Frequency Restoration Reserve Market. *Electric Power Systems Research* 185, 106399. doi: 10.1016/j.epsr.2020.106399

Schüller, Kristine, Aaslid, Per, Vereide, Kaspar and Skjelbred, Hans Ivar (2024). Practical Experiences from Application of a Comprehensive Simulator for Pumped Storage Hydropower Investment Decisions on a Real Investment Case. *2024 20th International Conference on the European Energy Market (EEM)*, 1–6. doi: 10.1109/eem60825.2024.10608922

Shapiro, Alexander (2011). Analysis of Stochastic Dual Dynamic Programming Method. *European Journal of Operational Research* 209(1), 63–72. doi: 10.1016/j.ejor.2010.08.007

Sioshansi, Ramteen, Denholm, Paul, Jenkin, Thomas and Weiss, Jurgen (2009). Estimating the Value of Electricity Storage in PJM: Arbitrage and Some Welfare Effects. *Energy Economics* 31(2), 269–277. doi: 10.1016/j.eneco.2008.10.005

Zisos, Athanasios, Sakki, Georgia-Konstantina and Efstratiadis, Andreas (2023). Mixing Renewable Energy with Pumped Hydropower Storage: Design Optimization under Uncertainty and Other Challenges. *Sustainability* 15(18), 13313. doi: 10.3390/su151813313